STRUCTURAL STEEL

ROBOSCAN FIT SHEET

Check a part against this sheet before anyone quotes it. Tick each box on the checklist. Even if every box is ticked, send a drawing or photo. CLOOS Americas confirms the fit before any quote.

WHAT IT CAN'T DO

01 Single wire only. No tandem, no TANDEM+, no torch change, no two-robot master and slave.

02 One robot per scanner.

03 It does not detect bevels, grooves or root gaps.

04 Material from 4 mm (0.16 in).

05 Tacks under 5 mm (0.20 in), never on the corners.

06 Square edges and corners.

07 Clean steel: no scale, rust, grease or oil. Ground or polished surfaces are excluded.

08 Nothing in the scan shadow. Only what the laser can see is found.

09 Controlled lighting, under 5000 lux.

10 The track must be longer than your longest part.

11 About 500 mm (20 in) before and 1000 mm (39 in) after a positioner is not welded.

12 Standard connections only. Combinations are confirmed case by case.

13 W-shape sections used in the US are checked against the library before any quote.

FIT CHECKLIST

- [ ] Material from 4 mm (0.16 in) for beams, columns and trusses. Sheet from 2 mm (0.08 in) and round bar from 5 mm (0.20 in) on gratings.
- [ ] Tacks under 5 mm (0.20 in), never on the corners. Tacks hold the gap and stop distortion.
- [ ] Square edges and corners. Rounded, deformed or welded over edges shift the measurement.
- [ ] Plates are rectangular and at least 5 mm (0.20 in) larger or smaller than the profile, tacked square to it.
- [ ] Clean steel, free of scale, rust, grease and oil. Ground or polished surfaces reflect and are excluded.
- [ ] No gaps or weld preparation. RoboScan does not detect bevels, grooves or root gaps.
- [ ] Nothing in the scan shadow. Only what the laser can see is found. Hidden joints need the part turned.
- [ ] Controlled lighting, under 5000 lux (465 foot-candles). Direct sunlight or a skylit bay is a problem.

PART

CHECKED BY

WHAT IT WELDS

BEAMS AND COLUMNS: 16 STANDARD CONNECTIONS

A1 Head plate	**A2** Column base	**B1** Reinforcing plate	**B2** Reinf. + cross plate
C1 Plate on web	**C2** Shear tab	**C3** Stiffener	**D1** Plate on profile
D2 Plate in profile	**D3** In profile, on flange	**E1** On flange, narrow	**E2** On flange, wide
E3 Lying, narrow	**E4** Lying, wide	**F1** Plate inside	**F2** Profile on profile

PROFILES AND SIZES

PROFILE	RANGE	INCHES
HEA, HEB, HEM	100 to 1000 mm	4 to 39 in
IPE	80 to 1000 mm	3 to 39 in
UNP channel	80 to 400 mm	3 to 16 in
Square and rectangular tube	20 to 250 mm	0.8 to 10 in
Angle	30 to 200 mm	1.2 to 8 in
Tee	T20 to T140	0.8 to 5.5 in

WELD SIZES

SIZE	LAYERS	AWS LEG
a3 to a6	Single layer fillet	3/16 to 5/16 in leg
a8 to a15	Two to six layers	7/16 to 13/16 in leg

- Profiles under 400 mm (16 in) are stood vertically on a plate.
- W-shape sections used in the US are checked against the library before any quote.
- The a size is the European throat. AWS gives fillets by leg size: leg = a x 1.41.

TRUSSES AND LATTICE GIRDERS

Top and bottom chords of I or H beam, with square tube diagonals between them. Rectangular plates are welded to the chords. It uses the same scanner and cell, with its own connection library.

RAILINGS, FENCES AND LADDERS

Upper and lower rails, and posts with a mounting plate. Infill can be vertical or horizontal, in flat bar, round, rectangular, square or edge set tube. Corner, tee, cross and through-pass joints. Infill angle within 45 degrees either way. Flat bar from 0.20 in (5 mm). Tube to tube is not possible.

GRATINGS AND MESH PANELS

A rectangular grid of bearing and cross bars with edge banding around it. Sheet from 0.08 in (2 mm) and round bar from 0.20 in (5 mm). Part IDs and sizes can come from your order database by barcode. All parts stand square to the panel.

SEND US A DRAWING OR PHOTO

CLOOS AMERICAS

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Videos and the full walkthrough: https://cloos-americas.github.io/roboscan-kit/